

IIT Foundation Module for Class VIII

Physics

- **Theme 1: The Language of Chemistry**

1. Symbol of an element; valency; formulae of radicals and formulae of compounds. Balancing of simple chemical equations.
2. Relative Atomic Masses (atomic weights) and Relative Molecular Masses (molecular weights)
 - Standard H atom or 1/12th of carbon-12 atom

- **Theme 2: Chemical Changes and Reactions**

1. Types of chemical changes
2. Energy changes in a chemical change

- **Theme 3: Water**

1. Water as a universal solvent
2. Hydrated and anhydrous substances
 - Hydrated substances
 - Anhydrous substances
 - Properties
3. Drying and Dehydrating Agents
4. Soft water and Hard water

- **Theme 4: Atomic Structure and Chemical Bonding**

1. Structure of an atom, mass number and atomic number, isotopes, octet rule
2. Electrovalent and covalent bonding, structures of compounds
 - Electrovalent Bond
 - Covalent Bond

- **Theme 5: The Periodic Table**

1. Dobereiner's Triads, Newland's Law of Octaves, Mendeleev's contributions

2. Modern Periodic Law, Modern Periodic Table (Groups and Periods)

○ **Theme 6: Study of the First Element – Hydrogen**

1. Position of Hydrogen in the periodic table and general group characteristics: valency electrons, burning, ion formation
2. Hydrogen from water, dilute acids, and alkalis
 - Hydrogen from water
 - Displacement from dilute acids
 - Displacement from alkalis
3. Preparation and collection of hydrogen (standard laboratory method other than electrolysis)
4. Industrial manufacture of hydrogen by Bosch process
5. Oxidation and reduction reactions

○ **Theme 7: Study of Gas Laws**

1. Behaviour of gases under changes in temperature and pressure; explanation in terms of molecular motion; Boyle's Law, Charles' Law, absolute zero, gas equation; simple calculations
2. Relationship between Kelvin and Celsius scales; standard temperature and pressure